

AAPT 2016 Conference: Climate Change

Day 3

NetCDF Tutorial



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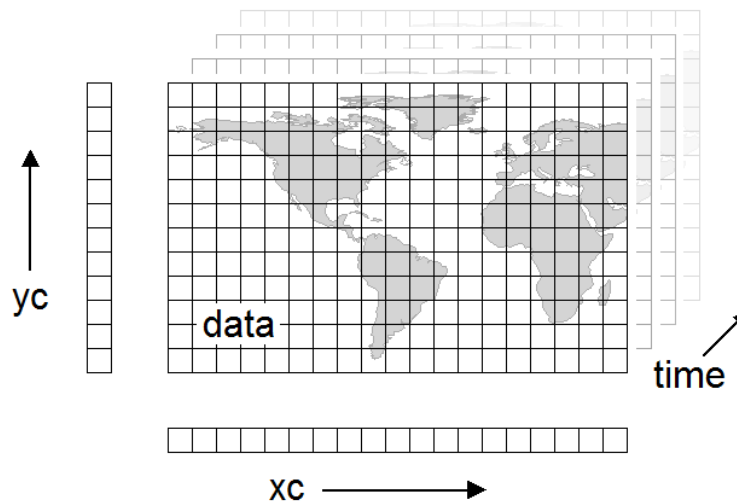
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About the netCDF DataSets

Network Common Data Form (netCDF) is a file format for storing multidimensional scientific data (variables) such as temperature, humidity, pressure, wind speed, and direction. You can use tools from the Multidimension Tools toolbox to create a raster layer, a feature layer, and a table view from a netCDF file. You can change the display by selecting a specific dimension value from the properties dialog box or by running the Select by Dimension geoprocessing tool. You can also perform analysis using a layer or table view created from a netCDF file.

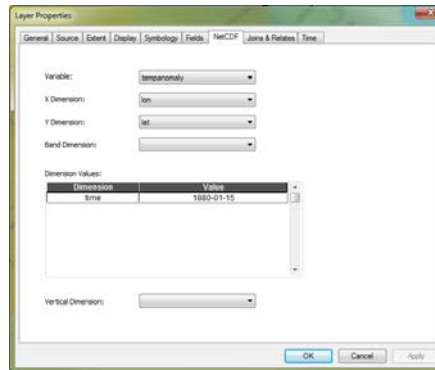
Most satellite data are time-stamped, meaning each data value is explicitly associated with a particular date and/or time. In the gridded data, time is usually defined at hourly, daily, and monthly resolutions, and each layer of the grid has one particular time, such as a particular month.

The first netCDF file that we will explore contains monthly data going back as far as January, 1880 (in some locations). Therefore, (April 2016 – January 1880) = 136 years and four months or 1636 months. Each month's value is one 2-dimensional data array, and the time dimension is an additional dimension. Thus, this netCDF formatted gridded temperature anomaly data is a 3-dimensional (1636x90x180) data array.



Data Set #1: Gridded Monthly Maps of Temperature Anomaly Data (NASA)

The first data set that we will examine originates from NASA's Goddard Institute of Space Studies (GISS) which is mandated to undertake broad studies of global changes looking at both natural and man-made changes in our environment that occur on a variety of time scales which impact the habitability of our planet. This would include such things as volcanic eruptions, El Niño and ice ages.



GISS Surface Temperature Analysis (GISTEMP) (From the GISS Surface Temperature Analysis (GISTEMP) Frequently Asked Questions (FAQ) <http://data.giss.nasa.gov/gistemp/FAQ.html>)

Temperature anomalies indicate how much warmer or colder it is than normal for a particular place and time. For the GISS analysis, normal always means the average over the 30-year period 1951-1980 for that place and time of year. This base period is specific to GISS, not universal. But note that trends do not depend on the choice of the base period: If the absolute temperature at a specific location is 2 degrees higher than a year ago, so is the corresponding temperature anomaly, no matter what base period is selected, since the normal temperature used as base point is the same for both years.

Note that regional mean anomalies (in particular global anomalies) are not computed from the current absolute mean and the 1951-80 mean for that region, but from station temperature anomalies. Finding absolute regional means encounters significant difficulties that create large uncertainties. This is why the GISS analysis deals with anomalies rather than absolute temperatures.

The GISTEMP analysis concerns only temperature anomalies, not absolute temperature. Temperature anomalies are computed relative to the base period 1951-1980. The reason to work with anomalies, rather than absolute temperature is that absolute temperature varies markedly in short distances, while monthly or annual temperature anomalies are representative of a much larger region. Indeed, we have shown (Hansen and Lebedeff, 1987) that temperature anomalies are strongly correlated out to distances of the order of 1000 km.

Graphs and tables are updated around the middle of every month using current data files from NOAA GHCN v3 (meteorological stations), ERSST v4 (ocean areas), and SCAR (Antarctic stations), combined as described in our December 2010 publication (Hansen et al. 2010). These updated files incorporate reports for the previous month and also late reports and corrections for earlier months.

The entire gridded surface air temperature anomaly data may download netcdf files containing selected series on a regular 2°x2° grid or the basic SBBX binary files. Note: These files are large. Compressed NetCDF files (regular 2x2 degree grid) <http://data.giss.nasa.gov/gistemp/>

Auxiliary Data: Means Based on Land-Surface Air Temperature Anomalies Only (Meteorological Station Data,)
Global-mean monthly, seasonal, and annual means, 1880-present, updated through most recent month: TXT , CSV

Data Set #2: NCEP/NCAR Reanalysis Monthly Means and Other Derived Variables. (NOAA)

From the Overview of the Physical Sciences Division (<http://www.esrl.noaa.gov/psd/about/>)

The NOAA/ESRL Physical Sciences Division (PSD) conducts weather and climate research to observe and understand Earth's physical environment, and to improve weather and climate predictions on global-to-local scales.

The NCEP/NCAR Reanalysis data set is a continually updated (1948–present) globally gridded data set that represents the state of the Earth's atmosphere, incorporating observations and numerical weather prediction (NWP) model output from 1948 to present. It is a joint product from the National Centers for Environmental Prediction (NCEP) and the National Center for Atmospheric Research (NCAR). (See also Meteorological reanalysis.)

The National Centers for Environmental Prediction (NCEP) and National Center for Atmospheric Research (NCAR) have cooperated in a project (denoted “reanalysis”) to produce a retroactive record of more than 50 years of global analyses of atmospheric fields in support of the needs of the research and climate monitoring communities.

Temporal Coverage:

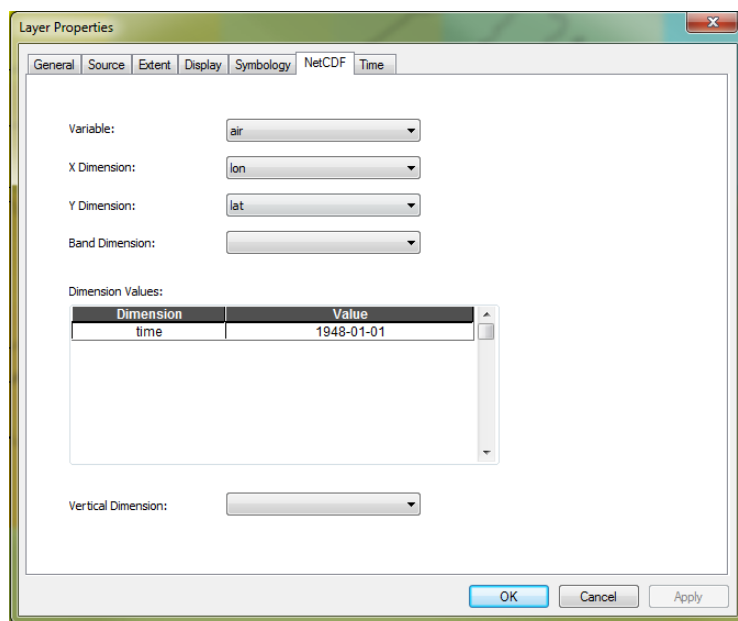
Daily and Monthly Values for 1948/01 - present.

Spatial Coverage:

2.5 degree latitude x 2.5 degree longitude global grid (144x73).

90N - 90S, 0E - 357.5E

<ftp://ftp.cdc.noaa.gov/Datasets/ncep.reanalysis.derived/pressure/air.mon.mean.nc>



About the netCDF Session

This Session

The first netCDF file that we will look at today will be the Gridded Monthly Maps of Temperature Anomaly Data (NASA). The three-dimensional variable `tempanomaly` represents the mean monthly temperature anomaly relative to a 30 year average from 1951-1980. In addition to lon and lat dimensions, the variable `tempanomaly` also has year, a third dimension to represent time steps.

This session is divided into four parts and is designed to allow you to explore the multidimension tools and the netCDF-related functionality in ArcGIS.

- In part 1, you will copy the tutorial data locally and create a raster layer from a netCDF file. You will then change the display by selecting a different time step and visualize the time dimension in ArcGIS.
- In Part 2 We will look at how define and visualize time dimension in ArcGIS
- In part 3, you will create a temperature time series table for a specific location.
- In part 4, you will create a model to find the temperature difference between two time steps.

Part 1: Displaying a raster layer from a netCDF file

This section investigates using the [Make NetCDF Raster Layer](#) tool and creating a raster layer using a variable from a netCDF file and then changing the display by selecting a different time step.

Complexity:
Beginner

Data Requirement:
ArcGIS Tutorial Data Setup

Data Path:
D:\AAPT\2016\ClimateChangeDay03\Source

Goal:
Create a raster layer from a netCDF file.

Copying tutorial data

Before working with multidimension tools, first copy the following data:

Steps:

1. Open Windows Explorer. You can use the shortcut key combination Windows logo key+E to open it.
2. Navigate to the M:\AAPT\2016 folder.
3. Right-click the ClimateChangeDay03 folder and click **Copy**.
4. Navigate to and right-click the D:\AAPT\2016 in Windows Explorer and click **Paste**.

Making a netCDF raster layer

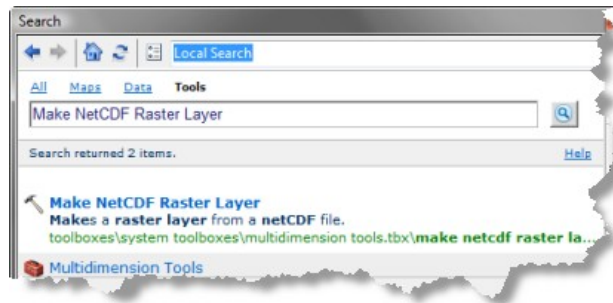
First we will run the [Make NetCDF Raster Layer](#) tool to create a raster layer using the tempranomaly variable from the gistemp1200_ERSSTv4.nc netCDF file.

Steps:

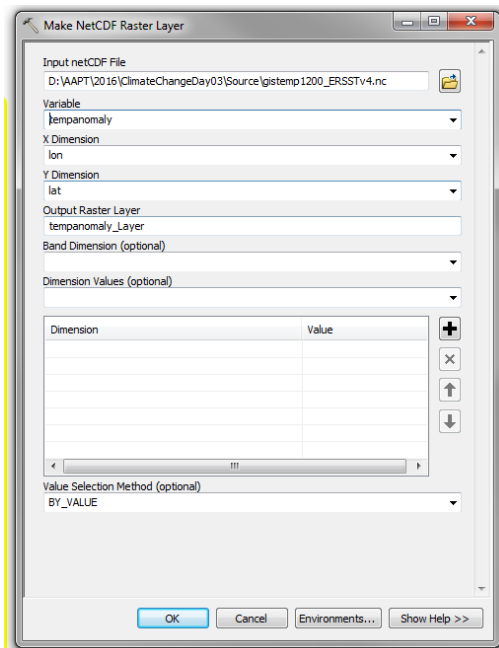
1. Start ArcMap by either double-clicking a shortcut installed on your desktop or using the All Programs list on the Start menu.
2. Click **OK** on the ArcMap startup dialog box.
3. Open the [Make NetCDF Raster Layer](#) tool.

 **Note:** Use the Search window to locate the [Make NetCDF Raster Layer](#) tool.

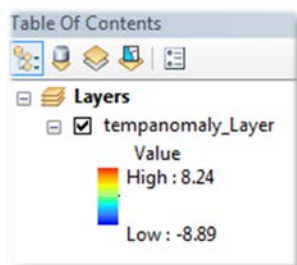
- a. Click **Search**
- b. Choose **Tools**.
- c. Type Make NetCDF Raster Layer.
- d. Click the magnifying glass icon.
- e. Click the Make NetCDF Raster Layer tool in the list.



4. For the **Input netCDF File** value, type or browse to D:\AAPT\2016\ClimateChangeDay03\Source\gistemp1200_ERSSTv4.nc.
5. Accept default values for **Variable** (tempanomaly), **X Dimension** (lon), and **Y Dimension** (lat) parameters.
6. Click **OK** to run the tool.



The tempanomaly raster layer is added to ArcMap. It displays the average monthly temperature anomaly of the year 1880, which is the default (first) time step. Note that the maximum and minimum temperature anomaly values in the table of contents are 8.24 and -8.89, respectively.

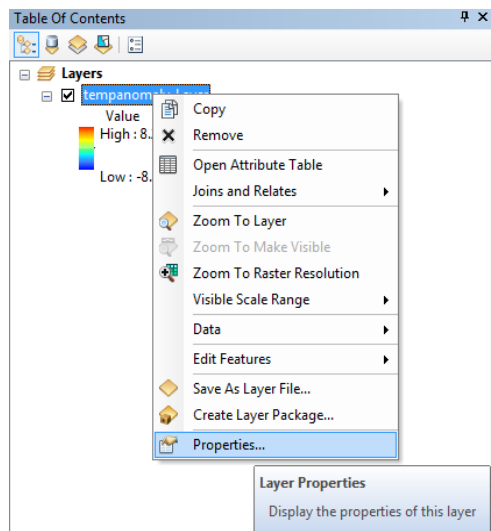


Displaying a specific time step

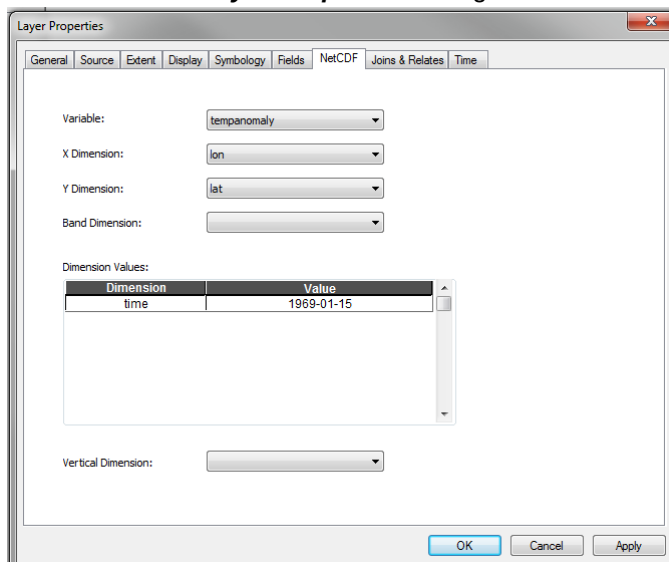
The raster layer is displayed using the default time step, which is 1880-01-15. You can change the display to show the temperature anomaly of January 1969 by changing the time step on the Layer Properties dialog box.

Steps:

1. Right-click the tempanomaly layer in the table of contents and click **Properties**.



2. Click the **NetCDF** tab.
3. Click the cell next to time under the **Value** column.
4. Click the arrow and choose 1969 - 01 - 15.
5. Click **OK** on the **Layer Properties** dialog box.



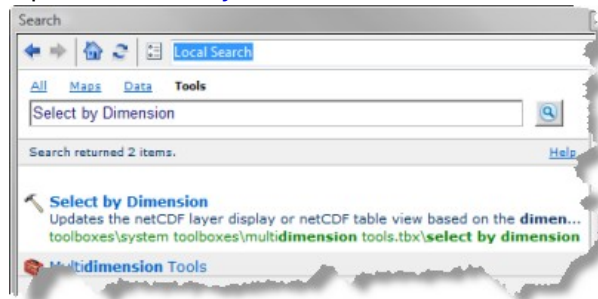
The tempanomaly layer now displays the temperature anomaly for January 1969. Note that the maximum and minimum temperature anomalies have changed to 7.11 and -11.42, respectively.

Returning to the default time step

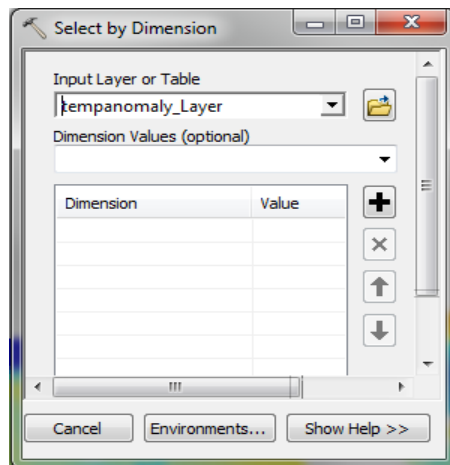
You can return to the default (the first) time step by changing the time dimension value on the Layer Properties dialog box, or you can run the [Select by Dimension](#) tool.

Steps:

1. Open the [Select by Dimension](#) tool.



2. For the **Input Layer or Table** value, choose `tempanomaly_Layer` from the drop-down list.
3. Accept default values for all other parameters.
4. Click **OK** to run the tool.

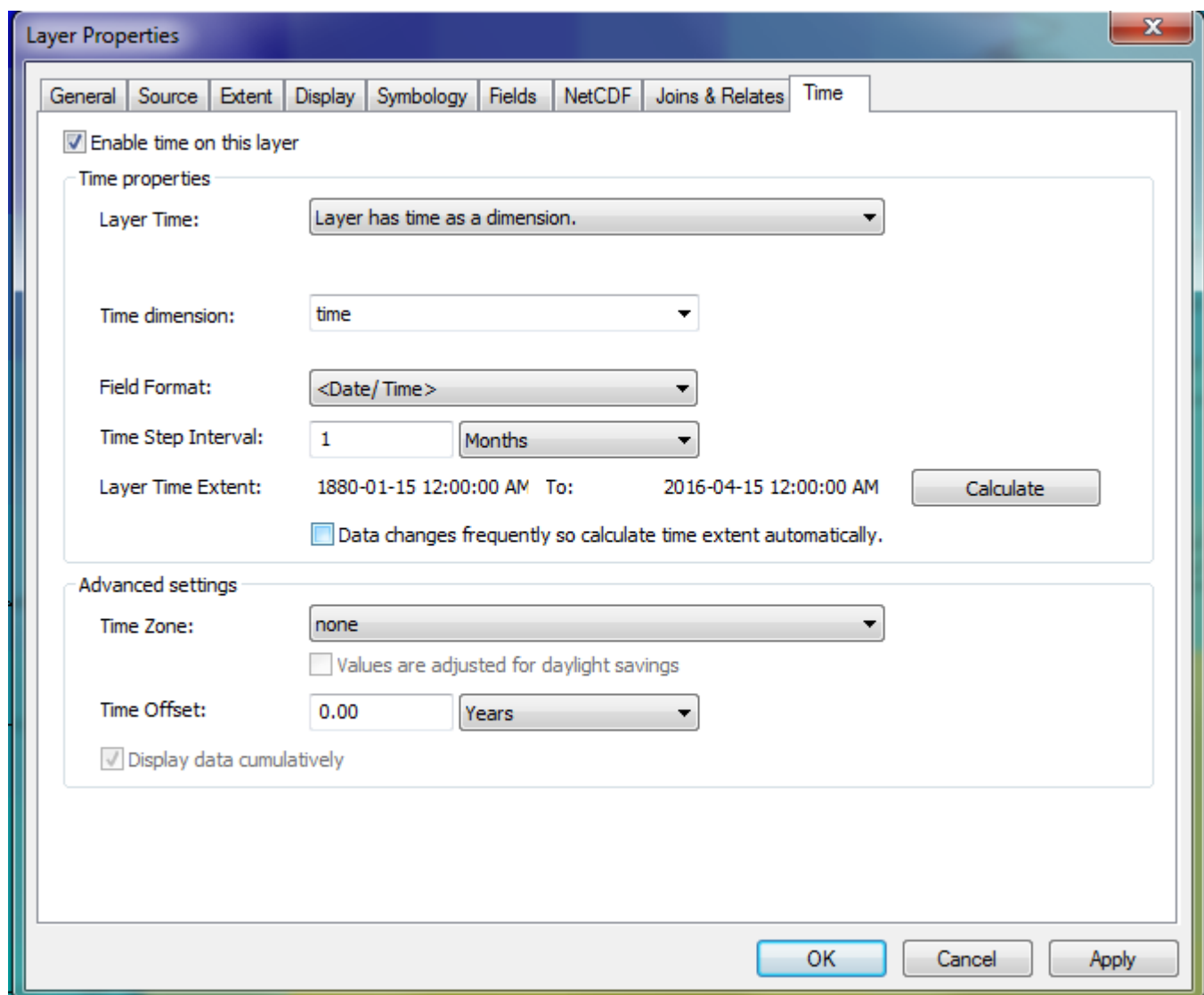


The display of the tempanomaly layer is back to the default time step (in other words, the temperature anomalies of January 1880). Note that the maximum and minimum temperature anomalies are now back to 8.24 and -8.89, respectively.

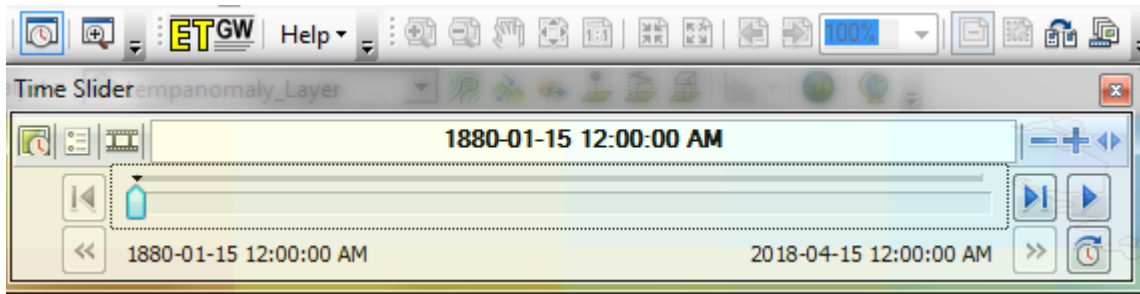
Part 2: Define and visualize time dimension in ArcGIS

Viewing the image at a time-of-interest:

- Right-click or double-click the imported layer named tempanomaly to bring out the “Properties” menu window.
- From the “Time” submenu, check the “Enable time on this layer” checkbox.
- Set the “Time dimension” as “time”, and set “Time Step Interval” to “1” (Month), as shown in the in the two highlighted red boxes in the image below.
- Click “OK” to close the window.
- You will now be able to visualize images of the time series temperature anomaly data using the “Time slider” tool in ArcMap. Click the “Time slider” button (the one with a little clock) to bring out the time player, and use the player to visualize the temperature anomaly data at different months.

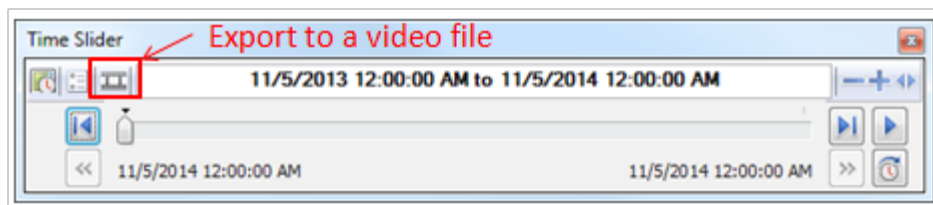


Enabling the time dimension in the temperature anomaly layer.



The time slider tool button and the time slider player.

- The time slider player also allows creating a visualization video file.



Export to a Video file.

Part 3: Creating a temperature table at a specific location

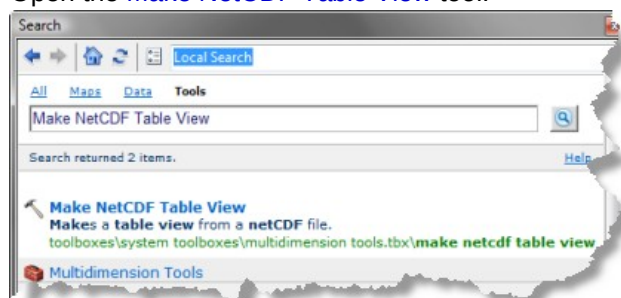
This part covers how to use the Make NetCDF Table View tool to create a table view using a variable from a netCDF file, then create a temperature profile from the table.

Making a netCDF table view

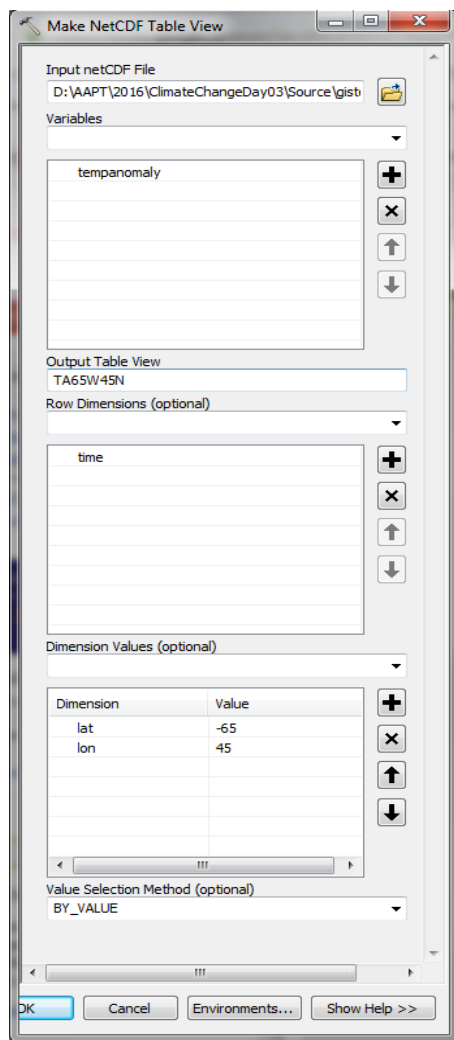
Run the Make NetCDF Table View tool to create a temperature time series table at a specific location using the tempanomaly variable from the gistemp1200_ERSSTv4.nc netCDF file.

Steps:

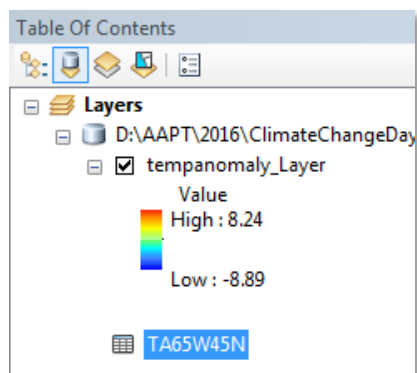
1. Open the [Make NetCDF Table View](#) tool.



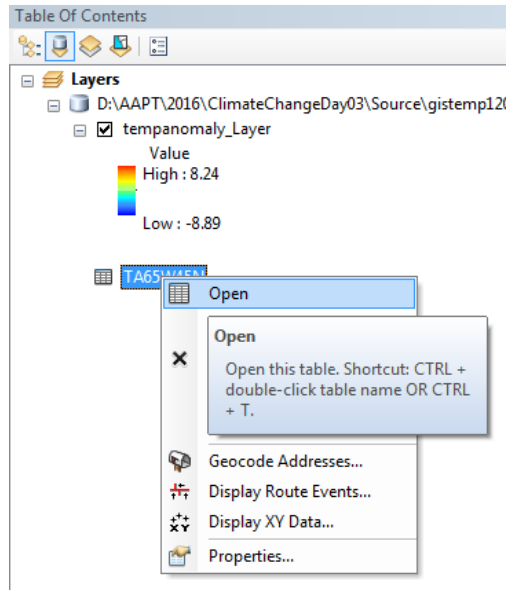
2. For the **Input netCDF File** value, type or browse to D:\AAPT\2016\ClimateChangeDay03\Source\gistemp1200_ERSSTv4.nc.
3. Click the **Variables** arrow and choose the tempanomaly variable.
4. Type TA65W45N for **Output Table View**.
5. Click the **Row Dimensions** arrow and choose the time dimension.
6. Click the **Dimension Values** arrow and choose the lon dimension.
7. Click the cell next to lon in the **Value** column and type -65.
8. Click the **Dimension Values** arrow again and choose the lat dimension.
9. Click the cell next to lat in the **Value** column and type 45.
10. Click **OK** to run the tool.



- Click the **List By Source** button in the table of contents to see the temperature anomaly table.



- Right-click temperature anomaly table in the table of contents and click **Open**.



13. Examine fields and values in the temperature anomaly table. N

Table			
TA65W45N			
	OID	time	tempanomaly
▶	1	1880-01-15	-0.35
▶	2	1880-02-15	-0.45
▶	3	1880-03-15	-0.5
▶	4	1880-04-15	-0.59
▶	5	1880-05-15	-0.56
▶	6	1881-01-15	-0.5
▶	7	1881-02-15	-0.5
▶	8	1881-03-15	-0.46
▶	9	1881-04-15	-0.56
▶	10	1881-05-15	-0.53
▶	11	1882-01-15	-0.57
▶	12	1882-02-15	-0.5
▶	13	1882-03-15	-0.47
▶	14	1882-04-15	-0.53
▶	15	1882-05-15	-0.59
▶	16	1883-01-15	-0.59
▶	17	1883-02-15	-0.63
▶	18	1883-03-15	-0.61
▶	19	1883-04-15	-0.65
▶	20	1883-05-15	-0.55
▶	21	1884-01-15	-0.52
▶	22	1884-02-15	-0.56
▶	23	1884-03-15	-0.49
▶	24	1884-04-15	-0.54
▶	25	1884-05-15	-0.58

14. Click the **List By Drawing Order** button in the table of contents.

On the main menu, click **File > Save**.

Part 4: Finding temperature difference between two time steps

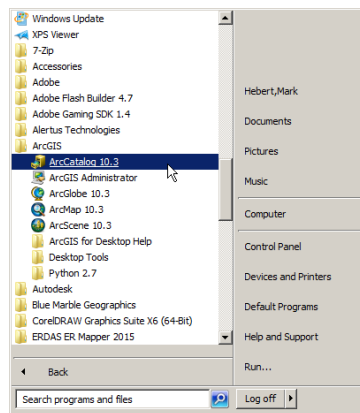
In this part we examine how to create a model to find the temperature difference between two time steps.

Creating a new toolbox

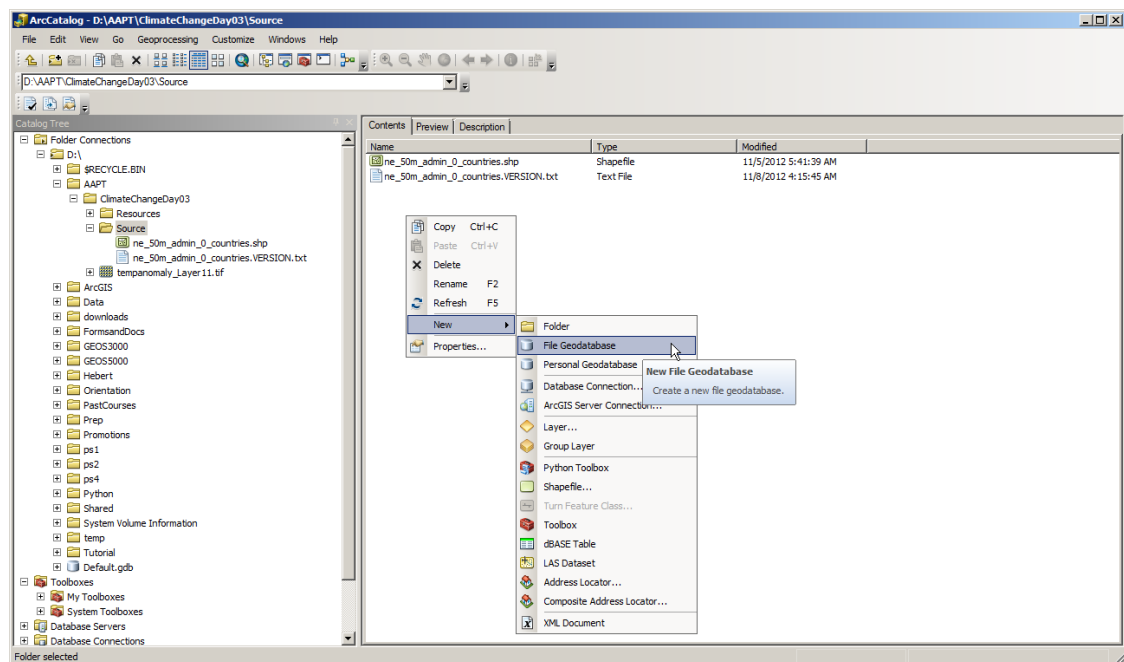
First create a new toolbox to store the model you will create in this exercise.

Steps:

1. Click the **ArcCatalog 10.3** icon from the start menu to open the **Catalog** window.

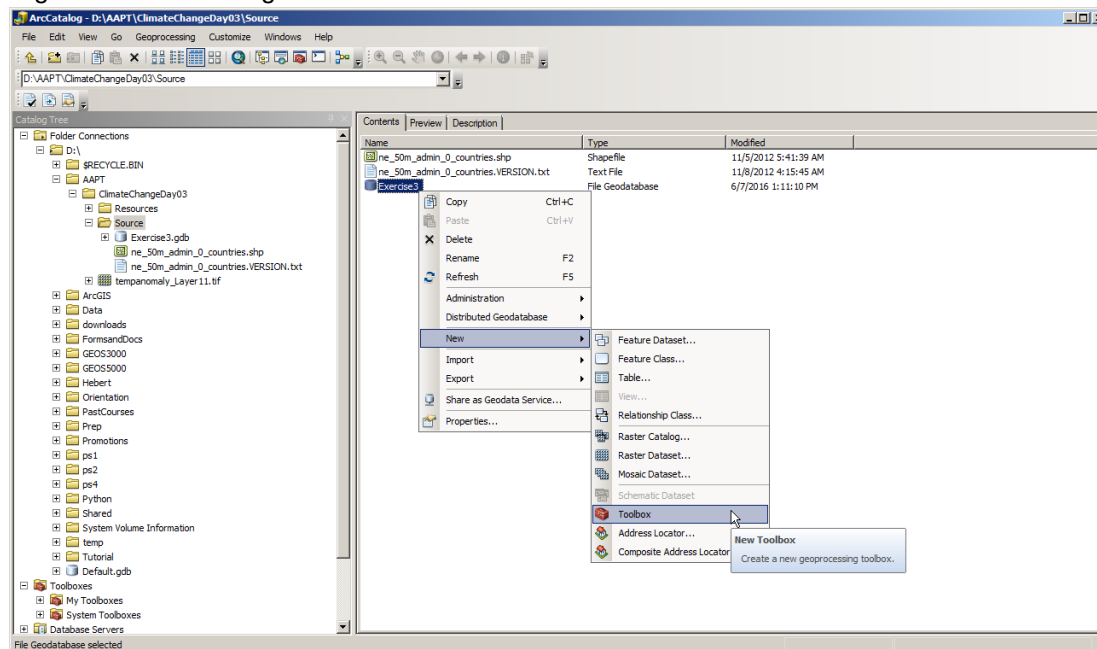


2. In Catalog, navigate to the D:\AAPT\2016\ClimateChangeDay03\Source folder. Right-click and select **New > File Geodatabase** in the popup menu.

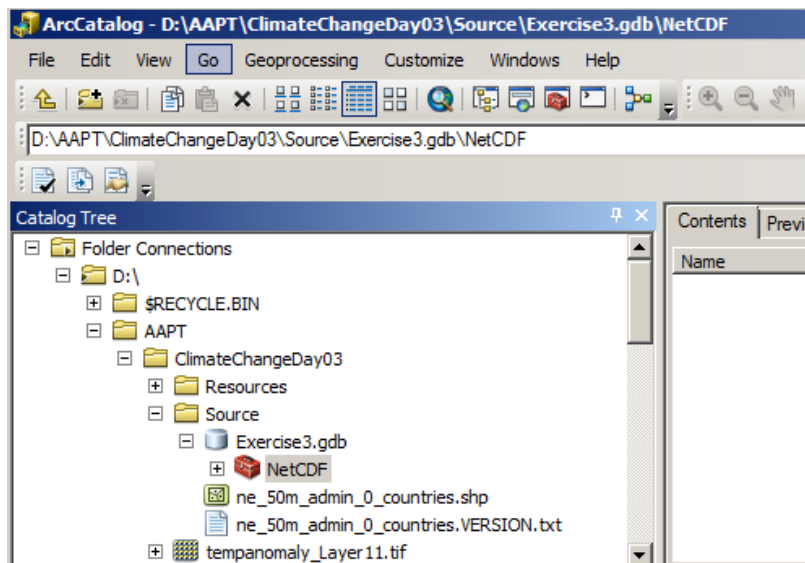


Name the Geodatabase **Exercise3**.

3. Right-click *Exercise3.gdb* and click **New > Toolbox**.



4. Change the name of the toolbox you just created from Toolbox to NetCDF.

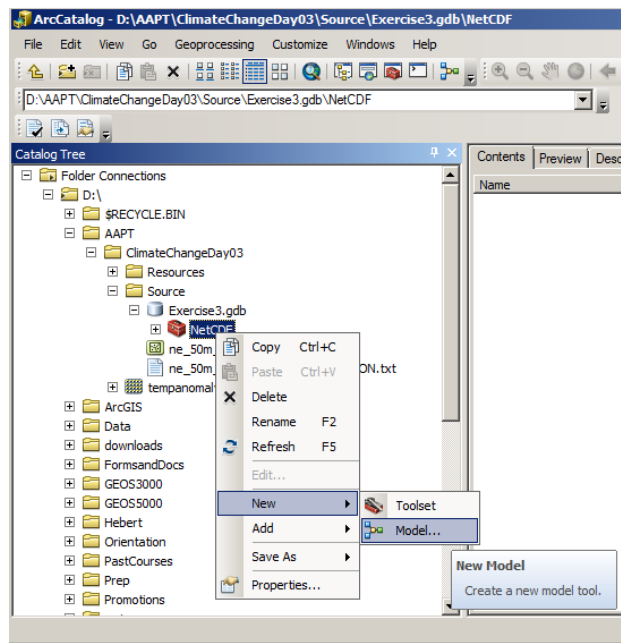


Creating a new model

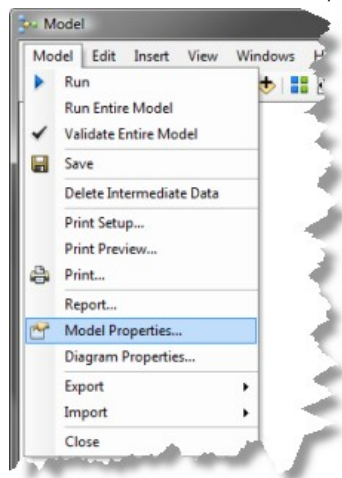
You'll create a new model and name it TemperatureDifference.

Steps:

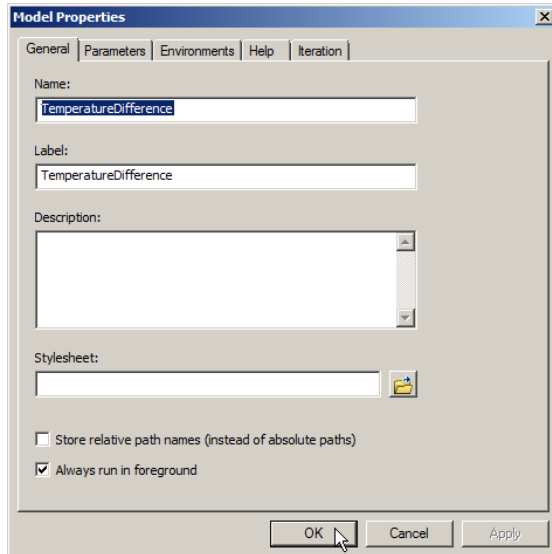
1. Right-click the NetCDF toolbox and click **New > Model**.



2. On the model's Main menu, click **Model > Model Properties**.



3. Click the **General** tab.
4. Type TemperatureDifference in the **Name** text box.
5. Type Temperature Difference in the **Label** text box.
6. Click **OK** to close the dialog box.

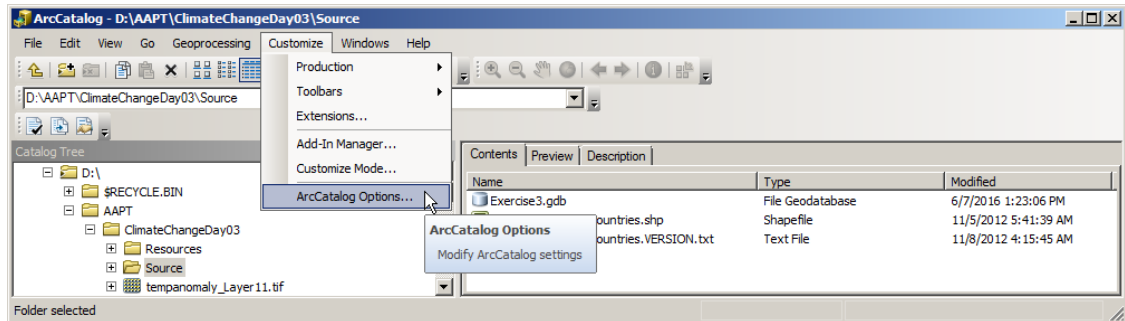


Customizing ArcCatalog to view NetCDF files.

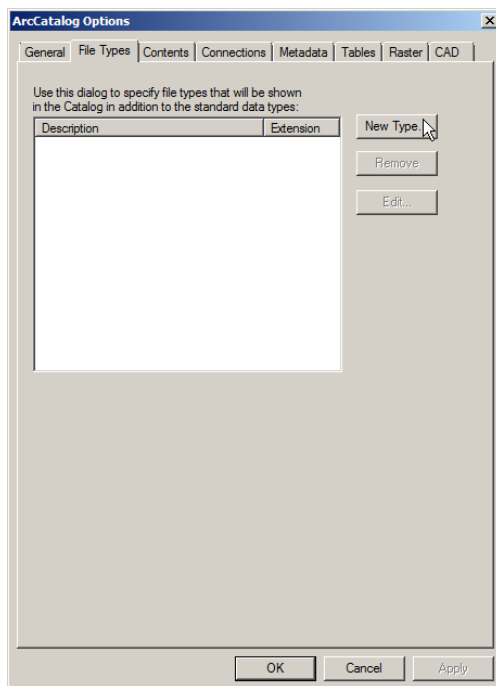
You'll customize ArcCatalog to view the netCDF files located in the Source folder.

Steps:

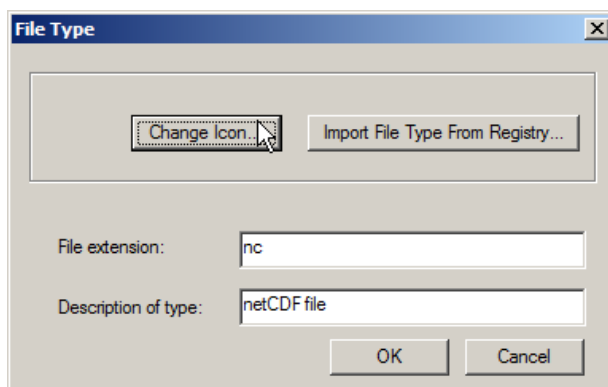
1. In the main ArcCatalog window choose **Customize > ArcCatalog Options**.



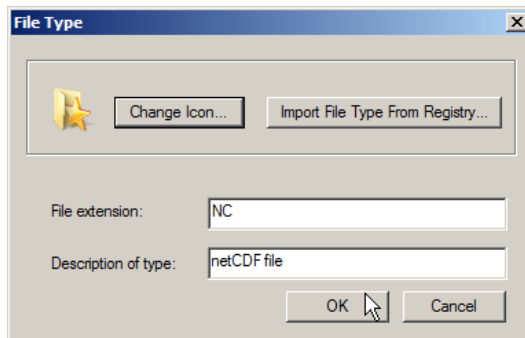
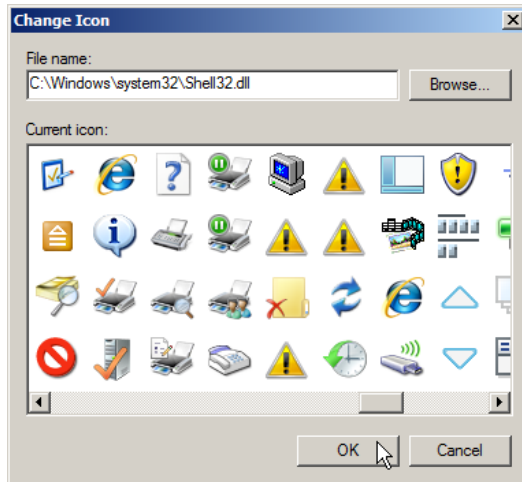
2. Click the **File Types** tab and **New Type**.



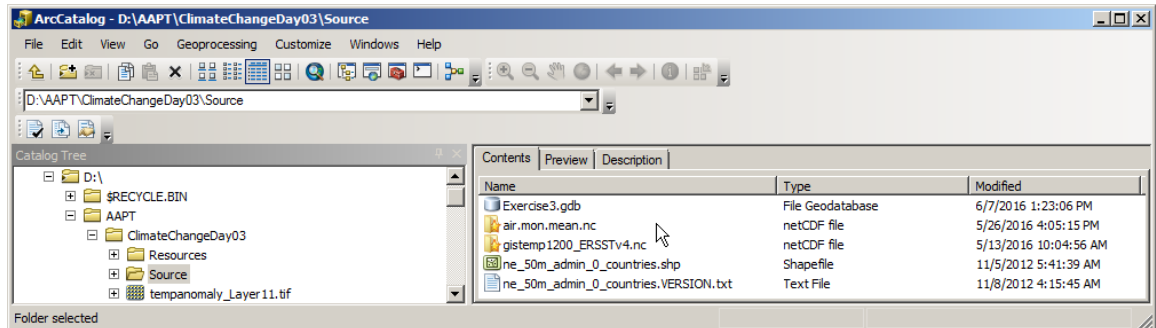
3. Enter nc as the **File extension** and **Description of type** netCDF file.



4. Pick **Change Icon** and choose one from the list and press **OK**.



5. Press **OK** to close the ArcCatalog Options dialog box and refresh ArcCatalog by pressing **F5**.
The **.nc** files should now be visible in ArcCatalog:

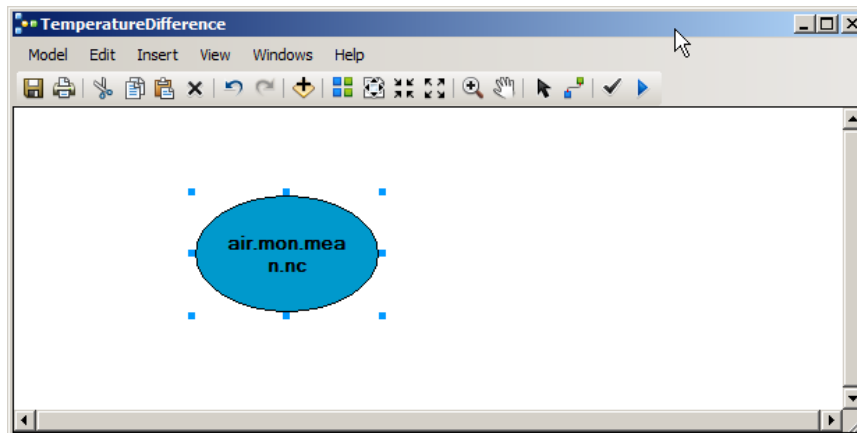


Building the Temperature Difference model

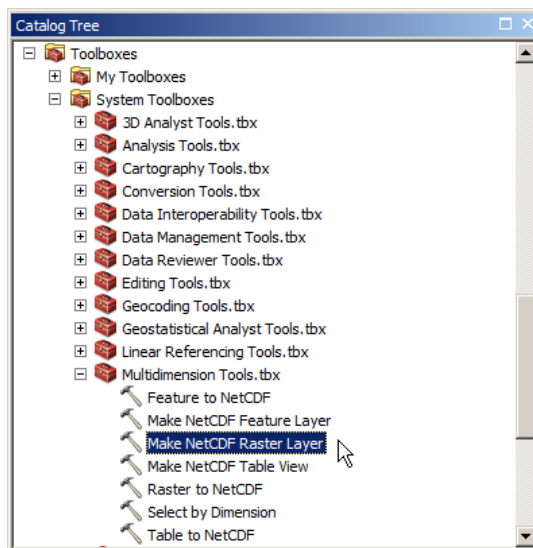
You'll add the netCDF file and Make NetCDF Raster Layer tool in your model to create two raster layers for the years 1951 and 2016.

Steps:

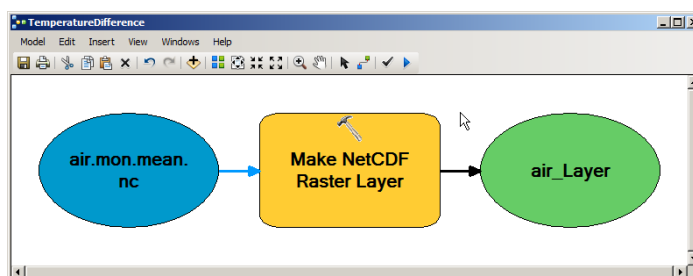
- From your Source folder in ArcCatalog, drag the **air.mon.mean.nc** file onto your model.



- Locate the [Make NetCDF Raster Layer](#) tool and drag it onto your model.



- Use the Connect button  to connect the **air.mon.mean.nc** variable as the input to the [Make NetCDF Raster Layer](#) tool.



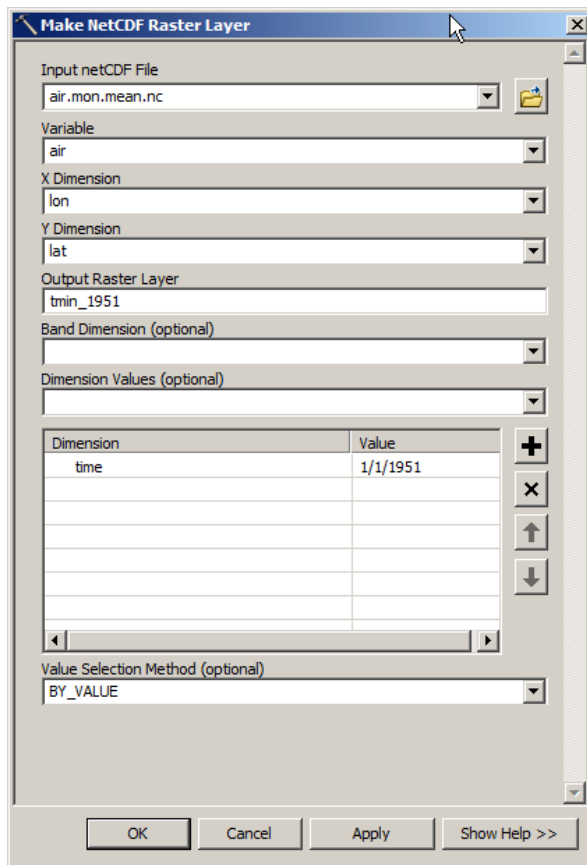
9. Double-click the [Make NetCDF Raster Layer](#) tool.
10. Type tmin_1951 in the **Output Raster Layer** text box.

What is a Raster Layer?

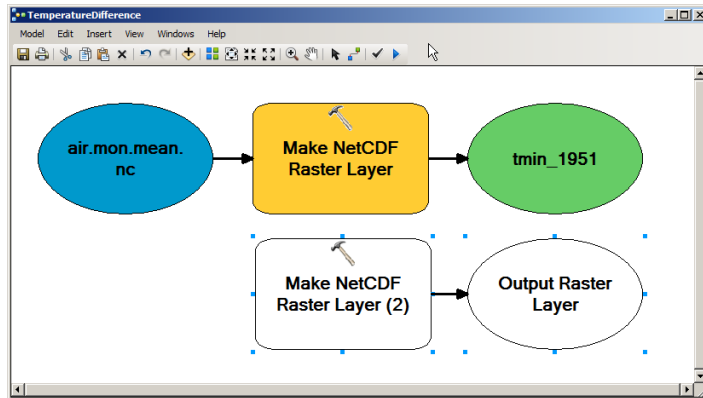
From ArcHelp: *The layer that is created by the tool is temporary and will not persist after the session ends unless the layer is saved to disk or the map document is saved.*

This tool can be used to make a temporary layer, so you can work with a specified subset of bands within a raster dataset.

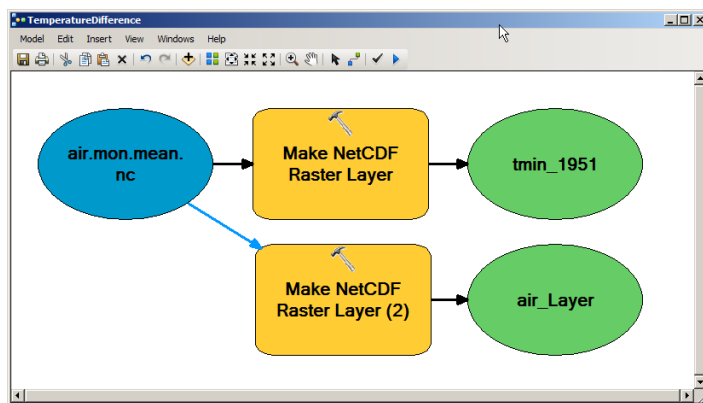
11. Click the **Dimension Values** arrow and choose the time dimension.
12. Click the cell next to year in the **Value** column, click the arrow, then choose 1/1/1951.
13. Click **OK** on the [Make NetCDF Raster Layer](#) dialog box.



14. Locate the [Make NetCDF Raster Layer](#) tool again and drag it onto your model.



15. Connect the `air.mon.mean.nc` variable to the [Make NetCDF Raster Layer \(2\)](#) tool.



16. Double-click the [Make NetCDF Raster Layer \(2\)](#) tool.
17. Type `tmin_2016` in the **Output Raster Layer** text box.
18. Click the **Dimension Values** arrow and choose the time dimension.
19. Click the cell next to year in the **Value** column, click the arrow, then choose 1/1/2016.
20. Click **OK** on the [Make NetCDF Raster Layer \(2\)](#) dialog box.

Make NetCDF Raster Layer (2)

Input netCDF File
 air.mon.mean.nc

Variable
 air

X Dimension
 lon

Y Dimension
 lat

Output Raster Layer
 tmin_2016

Band Dimension (optional)

Dimension Values (optional)

Dimension	Value
time	1/1/2016

Value Selection Method (optional)
 BY_VALUE

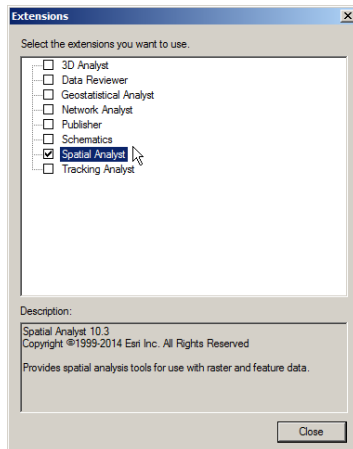
OK Cancel Apply Show Help >>

Finding temperature difference

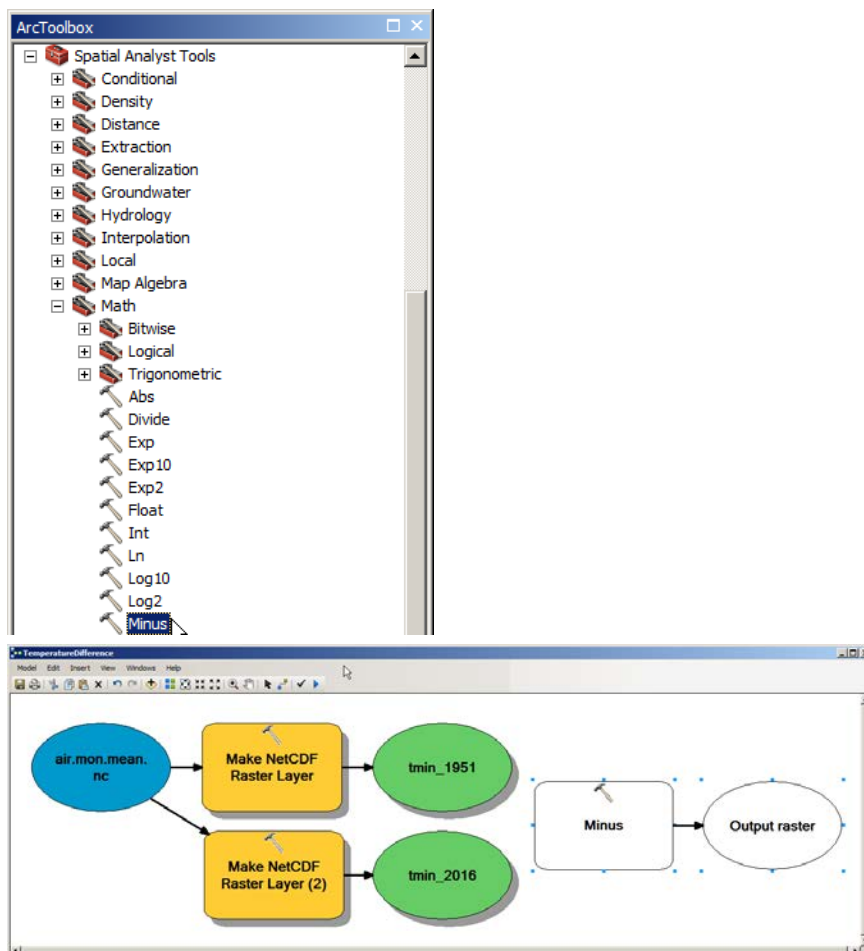
Now you'll finish up your model to find the minimum temperature difference between years 1951 and 2016 by subtracting two raster layers.

Steps:

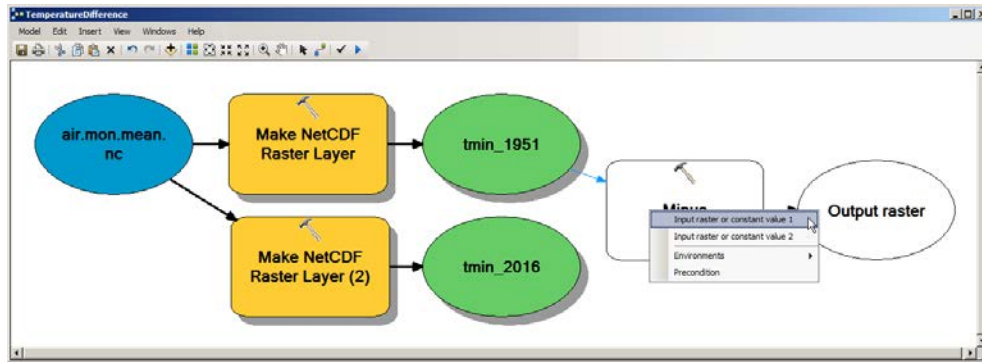
1. Be sure the Spatial Analyst Extension is turned on by choosing **Customize > Extension** and checking the Spatial Analyst.



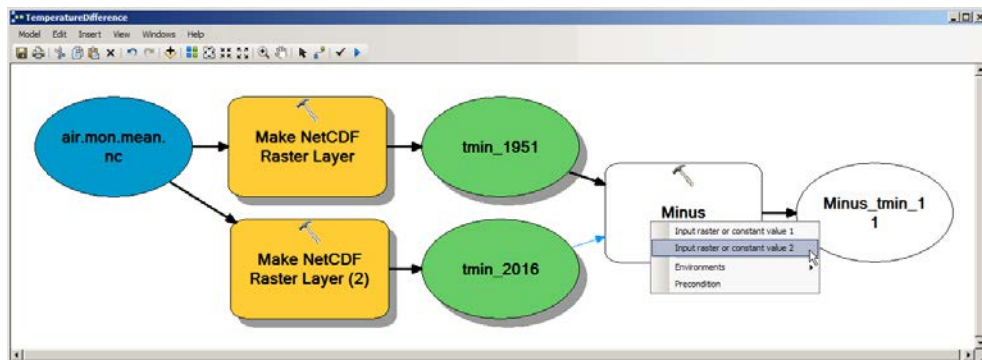
2. Locate the **Minus** tool in Spatial Analyst toolbox and drag it onto your model.



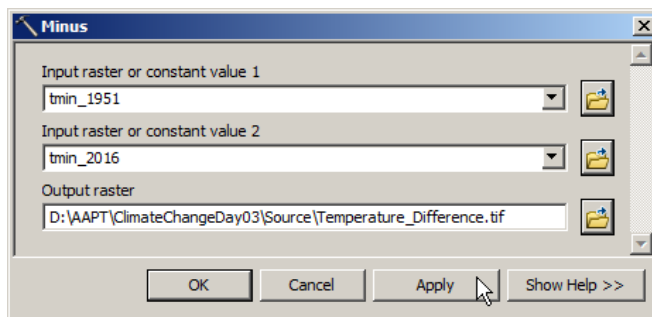
- Connect tmin_1951 to the **Minus** tool as the Input raster or constant value 1.



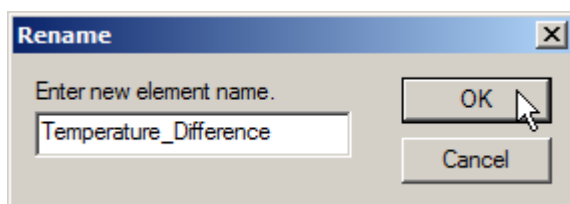
- Connect tmin_2016 to the **Minus** tool as the Input raster or constant value 2.



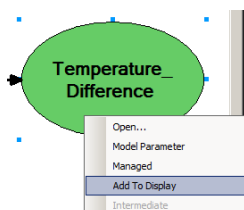
- Double-click the **Minus** tool and redirect the output to:
D:\AAPT\ClimateChangeDay03\Source\Temperature_Difference.tif



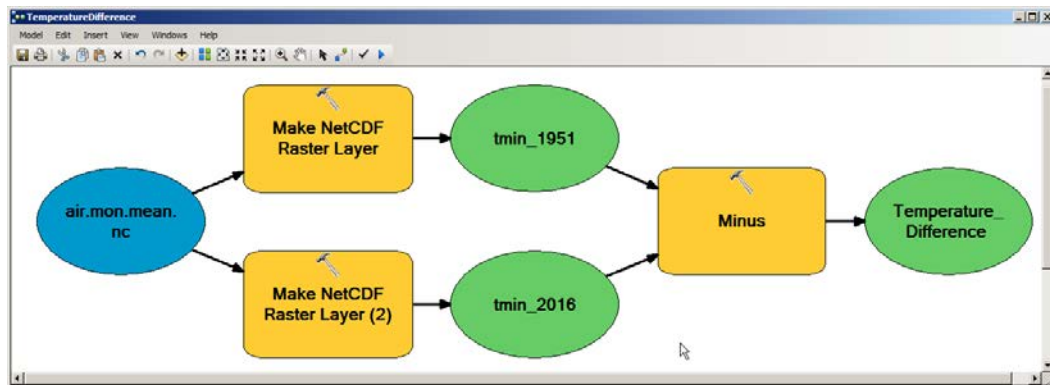
- Right-click the output of the **Minus** tool and click **Rename**.
- Type in the text box and click **OK**.



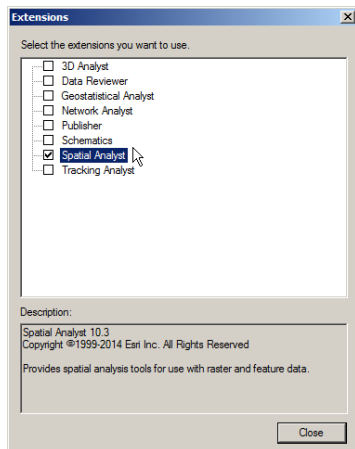
- Right-click Temperature_Difference and click **Add To Display**.



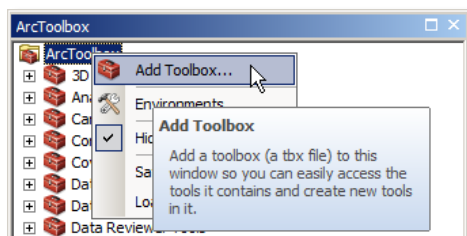
9. Click the **Auto Layout**  button to arrange the model elements.



10. Click the **Save** button  to save your model.
Close ArcCatalog and open ArcMap.
11. Be sure the Spatial Analyst Extension is turned on by choosing **Customize > Extension** and checking the Spatial Analyst.

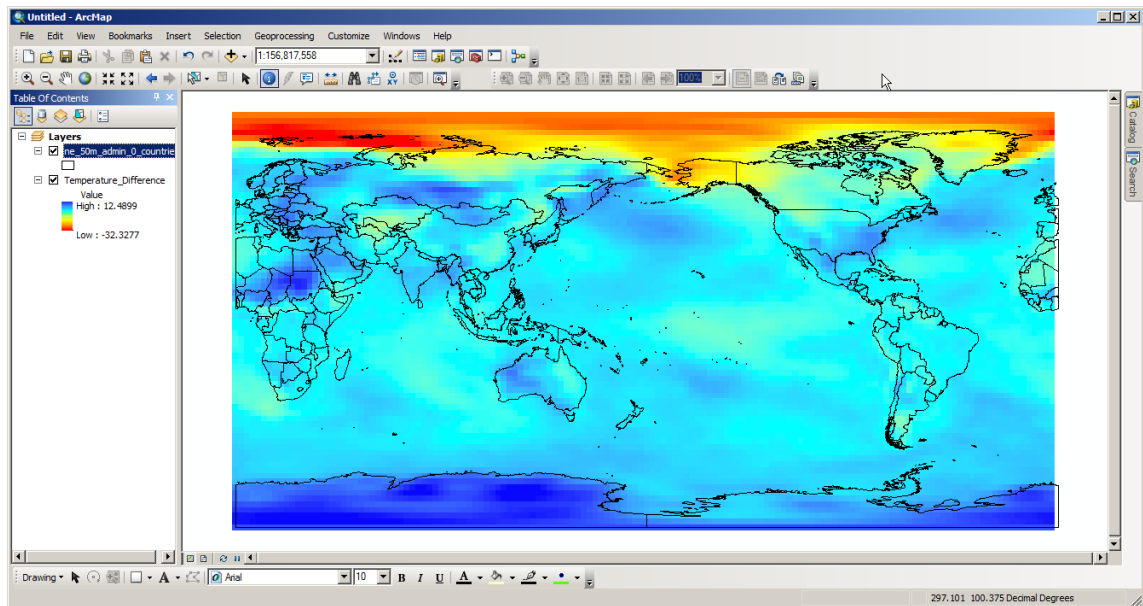
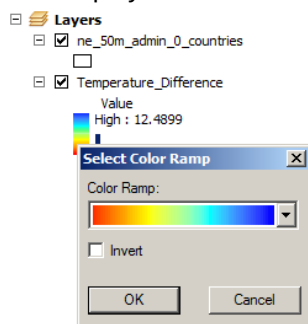


12. Add the model to the ArcToolBox in a new ArcMap session.



13. Click the **Run** button to run your model. The output raster layer is added to the table of contents.

14. Click the legend of the Temperature_Difference layer and choose a color ramp for better display.



Saving the map document

You are done with the exercise. However, you may want to save the map document to explore other functionalities.

Steps:

1. On the main menu, click **File > Save**.